Conductance Fluctuations in the SSH and Haldane Models

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***Abstract* -** The conductance properties of the Su-Schrieffer-Heeger (SSH) and Haldane models are studied numerically on Python using the Kwant package. Infinite leads are attached to two sides of the scattering region and electronic transports from one lead to another are computed numerically by Kwant. The conductance and electron density of the systems are plotted and visualized, showing resonance modes in both models as well as edge states. Next, random disorders are added to the systems and conductance fluctuations are calculated from a sample size of 500. It is found that the maximum fluctuation of the SSH chain occurs at shorter lengths and decays faster for larger disorders. The maximum fluctuations peaks at a value of ~0.28 e^2/h, independent of disorder magnitudes. Fluctuations at the estimated localization lengths of different disorder magnitudes also have root-mean-square values close to ~0.28 e^2/h but decreases as disorder magnitude increase. Universal conductance fluctuations for the Haldane model were not observed. Instead, fluctuations increase with increasing disorder magnitudes, this increase obeys a logarithmic function for small disorders.

Keywords – Mesoscopic, Electronic Transport, Conductance Fluctuation

1 INTRODUCTION

The physics of the conductance of a given conductor are well-established. However, as technology advances, smaller and smaller conductors can be fabricated and eventually, they no longer exhibit ohmic behaviors. Instead, quantum behaviors start to emerge. These conductors have length scales smaller than the mean free path of electron and the de Broglie wavelength and are known as mesoscopic. A mesoscopic lattice system is prone to disorders such as impurities during fabrication. These disorders scatter the transport of electrons through the system, causing fluctuations in the conductance and thus are also known as scatterers.

In this study, we will be investigating the conductance fluctuations of two of such disordered systems, the one-dimensional (1D) Su-Schrieffer-Heeger (SSH) model and the two-dimensional (2D) Haldane model. The SSH model was first used by Su, Schrieffer and Heeger in 1979 to model long chains of polyacetylene [1]. As for the Haldane model, it is conceived by Haldane in 1987 as a 2D system that exhibits quantum hall effect without the presence of a magnetic field [2]. Both models were chosen in this study as they behave as topological insulators which have unique properties that might prove to be promising in the fields of electronic, spintronic and quantum computing [3][4].

We will compute the conductance fluctuations of the two lattice models numerically on Python using Kwant, a free package for Python which aims to be easy to use yet powerful [5]. Multiple disordered samples will be generated randomly and the conductance for each will be computed. The conductance fluctuation is then simply the standard deviation of the conductance of all the samples. The study is conducted for different magnitudes of disorder and different lengths of the system for the SSH model. For the Haldane model, only the magnitudes of disorder will be varied as the size of the system will be fixed.

The focus of this study will be on the conductance fluctuations of the disordered SSH and Haldane lattice systems. We will not dive into the physics and theories of topological insulators, nor will other properties of the SSH and Haldane models be emphasized in detail. These includes energy band gaps, electrons probability density and topological edge states. However, they will be plotted and visualized to highlight important properties that might be reflected on the conductance fluctuations.

In this paper, we will first introduce the Python package Kwant by presenting the theory of conductance calculations of a mesoscopic conductor and showing how Kwant is able to build a similar system and does the calculations. Next, the structures and properties of the SSH model will be visualized and explained before presenting the results of its conductance fluctuations. Finally, the same process is done for the Haldane model before concluding this paper.

2 THEORETICAL FORMULATION

2.1 COMPUTING CONDUCTANCE

To measure the conductance of a given conductor experimentally, imagine placing the conductor in between two contacts and setting a potential difference between them, the conductance can be derived simply by measuring the current. Assuming that the conductor has a certain resistance, it is natural to think that if the length of the conductor gets shorter and closer to zero, the conductance gets larger and closer to infinity, with close to zero resistance. This is because we know that conductance is inversely proportional to the length of the conductor. However, experiments have shown that this is not the case [6][7]. Instead, the conductance approaches a limiting value of G_0, now known as the conductance quantum and has a value of 2 e^2/h. This limiting value comes from the resistance between the conductor and the contacts and thus is sometimes referred to as the contact resistance. In 1957, Rolf Landauer suggested the relation between conductance G and the overall transmission probability of electrons through the conductor [8]. This is the Landauer formula:

$$G = G_0 \Sigma_n T_n. \quad (1)$$

The transmission coefficient T_n describes the probability of an electron transmitting from one boundary of the conductor to the other. Therefore, the sum of all the transmission coefficients is the overall transmission probability of electrons through the conductor. Equation (1) shows that if this probability is one, then the conductance is simply the limiting value of G_0.

Extensions connecting the conductor to the contacts must be added, so that the boundary of the conductor where transmission coefficients are calculated from are not in contact with the contacts. This is so that contact resistances can be neglected in the transmission coefficients. These extensions are known as the leads as shown in Fig. 1a. The leads are assumed to have no resistance and are known as ballistic conductors.

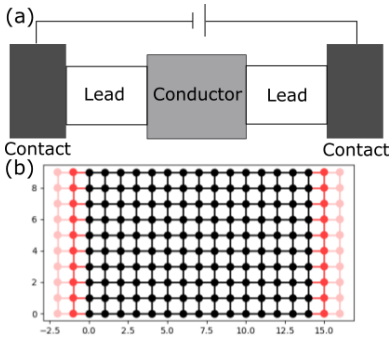

Figure 1 (a) Electron transports in experiments. (b) Simple 2D lattice built in Kwant.

Fig. 1b shows a simple 2D lattice system built in Kwant. The sites in red are the leads while the sites in black are the region to be investigated, known as the scattering region. From this point on, we will use the term scattering region to describe the region where transmission coefficients are calculated and thus is essentially the conductor that we are investigating.

Each scattering region has a scattering matrix and the transmission coefficients are the eigenvalues of that matrix. Kwant is capable of deriving the scattering matrix of a given system and solve for its transmission eigenvalues with just two lines of codes. This makes numerical calculations of conductance extremely easy and user-friendly with Kwant.

2.2 HAMILTONIAN AND BUILDING THE SYSTEM

For Kwant to be able to calculate the conductance of a scattering region, the lattice system must be built first. Using the tight-binding method, refined by Slater and Koster in 1954 [9], each lattice system can be described by a tight-binding Hamiltonian. For lattices with two sublattices like the SSH and the Haldane models, the Hamiltonian has the general form of

$$\boldsymbol{H} = t\Sigma_{r,i}\left(b_{r+i}^{\dagger}a_r + h.c.\right) + \Sigma_r\left(E_a a_r^{\dagger}a_r + E_b b_r^{\dagger}b_r\right). \quad (2)$$

Where t is the hopping term between sites, a and b are the sublattices operators and are either the creation or annihilation operators, i are the vectors directing the hopping and *h.c.* is the Hermitian conjugate. The second term is not important as the onsite potentials E_a and E_b are set to zero for non-disordered systems.

Kwant is able to generate the tight-binding Hamiltonian of a lattice system simply by declaring the sublattices, their relative position to each other, the primitive vectors of each sublattice and the hopping directions.

For the example lattice in Fig. 1b, there is only one sublattice and its primitive vectors are (1, 0) and (0, 1). The primitive vectors describe the position of another site of the same sublattice. Therefore, it has a Hamiltonian of

$$\boldsymbol{H} = t\Sigma_r\left(a_{r+x}^{\dagger}a_r + a_{r+y}^{\dagger}a_r + a_r^{\dagger}a_{r+x} + a_r^{\dagger}a_{r+y}\right). \quad (3)$$

Where x and y, the vectors directing the hopping, happens to be the primitive vectors (1, 0) and (0, 1). Leads are built the same way except a translational symmetry is declared to simulate an infinitely long lead. For the example lattice in Fig. 1b, the leads have a width of 9 unit cells and have a translational symmetry of 1 unit cell in the horizontal direction. The leads are then attached at two ends of the scattering region.

After building a lattice system on Kwant, it can be sent to various built-in solvers like the scattering matrix solver mentioned in the previous section. In this study, only the scattering matrix solver and the Hamiltonian eigenvalues solver are used to compute the conductance and to visualized the electrons probability density.

3 SSH MODEL

3.1 MODEL HAMILTONIAN

The SSH model is a 1D chain consisting of two sublattices in a unit cell. The hopping is staggered such that intracell and intercell hopping have different amplitudes t_1 and t_2. It has a tight-binding Hamiltonian of

$$\boldsymbol{H} = \Sigma_r\left[t_1\left(a_r^{\dagger}b_r + b_r^{\dagger}a_r\right) + t_2\left(b_r^{\dagger}a_{r+1} + a_{r+1}^{\dagger}b_r\right)\right]. \quad (4)$$

The SSH model has sublattices relative vector of (1, 0). This means that sublattice b is 1 unit to the right of sublattice a. Each sublattice has a primitive vector of (2, 0). This means that the next sublattice *a/b* will be 2 units to the right of the previous sublattice *a/b*. Therefore, the SSH chain will appear as (…*a-b-a-b-a-b*…). Each (*a-b*) is considered as one unit cell. Kwant builds the system by repeatedly placing a unit cell in the direction of the primitive vectors.

The SSH chain is built in Kwant simply by declaring these vectors for each sublattice, and declaring the hopping for a to b as t_1 (intracell) and the hopping for b to a as t_2 (intercell). By varying the values of the two hopping terms, energy band gaps in its dispersion relation can be opened, closed and varied.

Fig. 2a shows a SSH system of 50 unit cells plotted in Kwant, while Fig. 2b and Fig. 2c shows the conductance plot of a gapped system and a non-gapped system respectively. We set the hopping amplitude of the leads at 10 while the scattering region's hopping has values of 1 or smaller. The reason for the huge difference is to ensure that the input energy from the leads is larger than the band gap energy, no conductance would be seen if the input energy lies in the band gap energy.

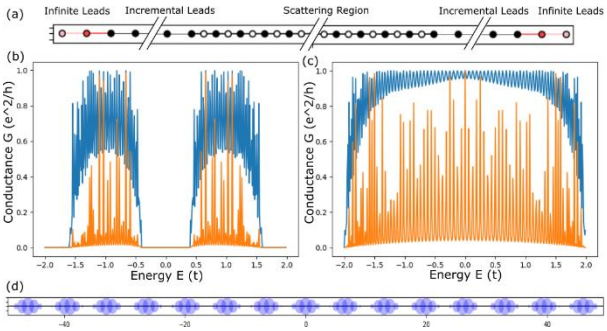

Figure 2 (a) SSH chain built in Kwant. (b) Conductance plot against energy of gapped system. (c) Conductance plot against energy of non-gapped system. (d) Electron density in the SSH chain.

However, due to the large difference in hopping amplitudes, there is an impedance mismatch between the system and the lead. This causes the conductance to be unstable which is shown as the orange plots of Fig. 2b and Fig. 2c. Therefore, we have added incremental leads between the scattering region and the leads. Each hopping between the incremental leads increment the hopping amplitude by a value of 0.5, until it reaches the leads' hopping value of 10. This smoothen the huge difference between the leads' and scattering region's hopping amplitudes and reduces impedance mismatch. The resulting conductance are shown as the blue plots in Fig. 2b and Fig. 2c.

In Fig. 2b, we set $t_1 = 0.6$ and $t_2 = 1$, while in Fig. 2c, $t_1 = t_2 = 1$. There will always be an energy band gap if t_1 is not equal to t_2. Fig. 2d shows the extended state of the system. The blue bubbles are the electron probability density of the system at a given energy. A resonant effect can be seen and is present at all energies that does not have zero conductance except for $E = 0$ in the non-gapped system, whereby the electron density is everywhere on the scattering region. This is due to the wave number $k = 0$ at $E = 0$ and thus the wavelength is infinite.

For the gapped system, $E = 0$ lies inside the band gap and hence there should be no conductance. However, electron density does not disappear completely and is at the left edge of the scattering region. If the values of t_1 and t_2 are swapped, such that now $t_1 = 1$ and $t_2 = 0.6$, the electron density is now at the right edge of the region. These two states are the edge states of the SSH system.

3.2 CONDUCTANCE FLUCTUATIONS

To compute the conductance fluctuations of a disordered SSH chain, we generated 500 non-gapped systems with all their onsite potentials randomized. This randomized onsite potential will serve as the disorders in the system. The randomization is generated from a uniform distribution and has a width of $[-d/2, d/2]$ where d is the disorder magnitude and is varied from 0.3 to 0.7. The fluctuation σG is computed as the standard deviation of the conductance at $E = 0$ of the 500 systems.

$$\sigma G = \sqrt{< G^2 > - < G >} \tag{5}$$

The fluctuations should be the same regardless of the chosen energy. This is done for chains with lengths of 10 to 500 unit cells with an increment of 5 unit cells. Fig. 3a and 3b shows the fluctuations and natural log of the average conductance as a function of the unit cell length L.

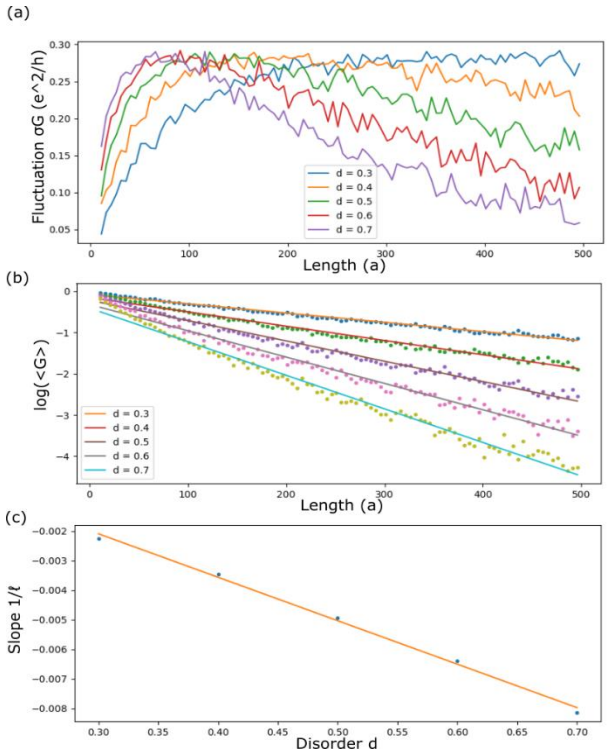

Figure 3 (a) Plot of conductance fluctuation against length of SSH chain. (b) Plot of log(<G>) against length of SSH chain. (c) Plot of $1/\ell$ against disorder of SSH chain.

We found that the conductance fluctuations peaks at shorter chain lengths as disorder magnitude increases as shown in Fig. 3a. The decay of the conductance fluctuation is also faster for increasing disorder magnitudes. From Fig. 3b, we see that the average conductance <G> has an approximate exponential relation with the system length L where ℓ is the localization length:

$$\log(< G >) = \log(A) - L/\ell, \tag{6}$$

$$< G > = Ae^{-\frac{L}{\ell}}. \tag{7}$$

Fig. 3c shows the plot of $1/\ell$ against disorder magnitudes d. We found that it has a relation of

$$1/\ell = -0.0147d + 0.0023. \quad (8)$$

From (7), we approximated the localization lengths for each disorder magnitudes and compute the conductance fluctuations as a function of energy at these localization lengths with their associated disorder magnitudes. This is shown in Fig. 4 with their respective RMS values in Table 1 and plotted as the orange dashed lines.

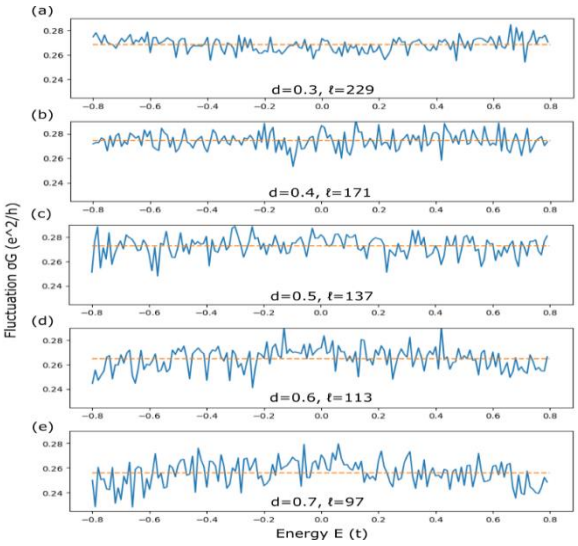

Figure 4 Conductance fluctuations of SSH chain at different disorders and localization lengths.

Table 1 RMS values of σG for SSH model

d	ℓ	RMS(σG)
0.3	229	0.269
0.4	171	0.275
0.5	137	0.273
0.6	113	0.265
0.7	97	0.256
Ave RMS(σG)		0.267 ± 0.007

We can see that the conductance fluctuations have a larger variation for larger disorder magnitudes. We would expect the RMS values of all the conductance fluctuations at the localization lengths to close to the maximum fluctuation value. However, the RMS values decrease for larger disorder magnitudes. This shows that the exponential relation between <G> and L might not be as accurate for larger disorders since the relation between the localization lengths and disorders were approximated from it.

The maximum fluctuations for all disorders does not exceed ~0.28 e^2/h, this value is close to a similar study by Gangopadhyay and Sen in 1992 where they found that the conductance peaks at ~0.3 e^2/h for a 1D system regardless of disorder widths [10].

4 HALDANE MODEL

4.1 MODEL HAMILTONIAN

The Haldane model is a 2D honeycomb lattice with two sublattices per unit cell, one on top of another with relative vector (0, 1). It has the following tight-binding Hamiltonian:

$$\boldsymbol{H} = t_1 \Sigma_{r,i}\left(b_{r+i}^{\dagger} a_r + h.c.\right) + t_2 \Sigma_{r,j}\left(a_{r+j}^{\dagger} a_r + b_{r+j}^{\dagger} b_r + h.c.\right) \quad (9)$$

The nearest-neighbor hopping term t_1 = 1 and i are the three vectors pointing to a site's nearest neighbors. Different from the SSH model, the Haldane model features a next-nearest-neighbor hopping with hopping term t_2 = 0.1. Similar to the nearest-neighbor hopping, the vectors j points to the next-nearest-neighbors' positions and happens to be the primitive vectors (1, 0) and (√3/2, 0.5).

We built the scattering region to be investigated with a size of 23 by 23 unit cells. It is shaped as shown in Fig. 5a as the system is built with an array of unit cells along the primitive vectors, this causes the scattering region to slant along (√3/2, 0.5). Like the SSH system, we have added incremental leads between the scattering region and the infinite leads as shown in Fig. 5a due to impedance mismatch. In Fig. 5b it can be seen how the incremental leads are attached to the scattering region, as well as the honeycomb lattice structures. Fig. 5c shows the plot of conductance of the scattering region against energy. The blue and orange plots are the conductance with and without incremental leads. Similar to the SSH system, adding incremental leads assures a larger and more stable conductance.

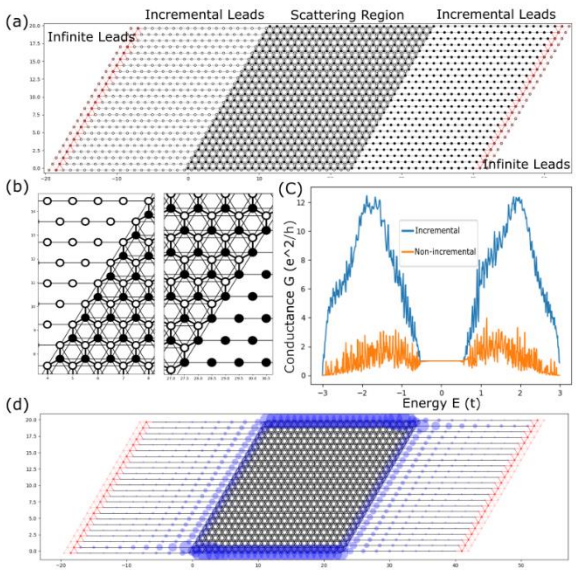

Figure 5 (a) Haldane model built in Kwant. (b) Close-up of structures of lattice. (c) Conductance against energy plot of Haldane lattice. (d) Edge states visualized by electron density.

We also note that in Fig. 5c, between $E \sim -0.6$ and $E \sim 0.6$, the conductance is stable at exactly 1 e^2/h. This is the edge state of the system. Unlike the SSH system, the edge states connect the band gap and have non-zero conductance. For edge states, the electron probability densities are at the edge of the scattering region as visualized in Fig. 5d.

4.2 CONDUCTANCE FLUCTUATIONS

To compute the conductance fluctuations of disordered Haldane model, again we will generate 500 samples with all its onsite potential randomized. Unlike the study of the SSH model, the size of the scattering region is not varied and fixed at 23 by 23 unit cells. Instead, the conductance fluctuations and average conductance are calculated at a range of energies of [-3, 3] and plotted as shown in Fig. 6a and 6b.

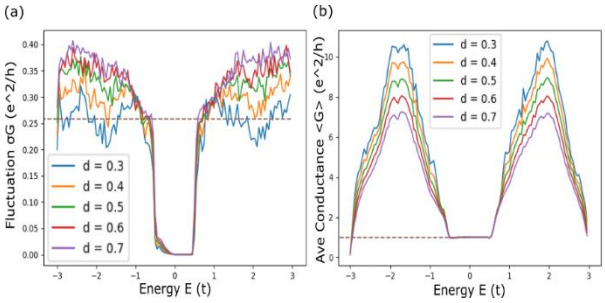

Figure 6 (a) Plot of conductance fluctuations against energy of Haldane model. (b) Plot of average conductance against energy of Haldane model.

We see that conductance fluctuations at the edge state energies are zero and the average conductance are also fixed at 1 e^2/h (dashed line), regardless of disorder magnitudes. This is what we would expect from the edge states. It is obvious to see that the average conductance decreases as disorder magnitude increases as shown in Fig. 6b. However, in Fig. 6a, it is seen that conductance fluctuations increases with increasing disorder magnitudes, and universal conductance fluctuations of ~0.258 e^2/h (dashed line) were not observed, although the fluctuations for $d = 0.3$ were close with a RMS value of ~0.261 e^2/h. The conductance fluctuations for different disorder magnitudes seems to be approximately the same from $E \sim -1$ to $E \sim 1$, a little larger than the edge state energies. But starts to diverge from each other for energies not within that range.

We also found that the RMS values of the conductance fluctuations obeys a logarithmic growth for disorder magnitudes smaller than 0.5, the relation does not hold and starts to diverge for larger disorder magnitudes as shown in Fig. 7. Table 2 shows the RMS values of the conductance fluctuations at energies where they diverge.

Table 2 RMS values of σ*G* for Haldane model

d	RMS(σ*G*)
0.3	0.261
0.4	0.302
0.5	0.331
0.6	0.350
0.7	0.363

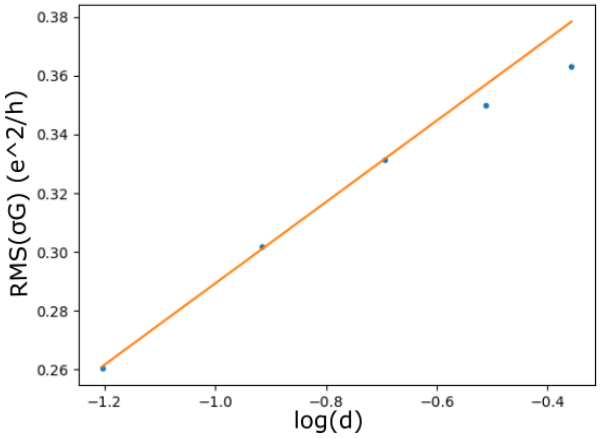

Figure 7 Plot of RMS of conductance fluctuations against natural log of disorder magnitudes.

For disorder magnitudes smaller than 0.5, the linear relation in Fig. 7 has slope of 1/log(*B*) and intercept of *C* where *B* is the base of the logarithm relation:

$$RMS(\sigma G) = \log(d) / \log(B) + C \qquad (10)$$

$$RMS(\sigma G) = \log_{1359} d + 0.427888 \qquad (11)$$

Equation (10) holds for all disorder magnitudes smaller than 0.5.

4 CONCLUSION

In summary, we are able to build the SSH and Haldane models in Kwant easily by declaring a few features of the systems. Next, we can attach leads and solve for the conductance and energy eigenvalues of the system using the Landauer formula, allowing us to observed band gaps and edge states in both systems. We proceed to randomly generate 500 samples of disordered systems and calculate the conductance fluctuations as a function of energy for both systems, and a function of system length for the SSH model. We found that the maximum conductance fluctuations for the SSH model peaks at ~0.28 e^2/h regardless of disorder magnitudes, and that the RMS values of the fluctuations at the localization lengths are also close to this value. For the Haldane model, we found that there are no conductance fluctuations at the edge state energies and that the fluctuations increases for increasing disorder magnitudes outside of the edge states. We found that this increase obeys a logarithmic relation with the disorder as shown in (10) for disorders magnitudes smaller than 0.5.

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